THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM THREE ANNUAL EXAMINATION
032 CHEMISTRY

Time: 3 Hours **November 2025**

Instructions

1. This paper consists of sections **A**, **B** and **C** with a total of eleven questions.
2. Answer all questions in sections **A** and **B** and only two questions from section **C**.
3. Non-programmable calculators are allowed in the examination room.
4. Write your examination number on every page of your answer booklet(s)
5. The following constants may be used;

   Atomic masses;

   H=1, O=16, C=12, Cu=63.5, S=32, Cl=35.5, Ca=40, Na=23, N=14, Fe=56, Ag=108, Br=80, Zn=65

   Avogadro's constant = 6.02 x 10^{23} G.M.V at S.T.P = 22.4 dm^3

   1 Faraday =96,500 coulombs

   Standard pressure = 760mmHg

   Standard temperature = 273K

   Density of water = 1000kg/m3

   1 litre = 1dm3=1000cm3

   Specific heat capacity of water = 4.18KJ/KgK$^-$

SECTION A (16 MARKS)

Answer all questions in this section.

1. For each of the items (i)–(x), choose the correct answer from the given alternatives and write its letter beside the item number in your answer booklet.

   (i) The number of neutrons in an atom of ^{13}C is:

   A. 6 B. 7 C. 13 D. 19 E. 21

   (ii) Which of the following is an example of a physical change?

   A. Burning a piece of paper.

B. Rusting of iron.
C. Melting of ice.
D. Decomposition of hydrogen peroxide.
E. Explosion of a bomb

(iii) The number of moles in 49 g of sulphuric acid (H_2SO_4) is:
A. 0.5 moles
B. 1.0 moles
C. 2.0 moles
D. 4.0 moles
E. 0.6 moles

(iv) The reaction between an acid and a base is called:
A. Oxidation
B. Reduction
C. Neutralization
D. Hydrolysis
E. Syntheses

(v) Which of the following elements belongs to Group II and Period 3 of the periodic table?
A. Mg B. Ca C. Al D. Na Fe

(vi) The chemical formular for manganese (IV)oxide is…
A. MnO_4
B. MnO_3
C. MnO_2
D. Mn_2O_3
E. Mn_4O_2

(vii) During a laboratory experiment at Seypalm secondary school, a student measured a charge of 1 coulomb after allowing a current to flow through a certain point in an electrolyte for 0.5 seconds. What was the current that flowed for the 0.5seconds?
A. 1 ampere
B. 2.5 amperes
C. 0.5 amperes
D. 2 amperes
E. 1.5 amperes.

(viii) A solution with a pH of 8 is considered:
A. Acidic
B. Basic
C. Neutral
D. Very acidic
E. Amphoteric

(ix) In the electrolytic refining of copper, the impure copper is made the:
A. Anode
B. Cathode

C. Electrolyte
D. Salt bridge
E. Cation

(x) A student added a few drops of barium chloride in a test tube containing unknown sample solution; a white precipitate was observed. The unknown sample is most likely to be:

A. Metal carbonates
B. Metal chloride
C. Metal sulphate
D. Metal nitrate
E. Metal hydrogen carbonate

2. Match the items in **List A** with their corresponding descriptions in **List B**.

LIST A	LIST B
(i) Sublimation (ii) Isotopes (iii)Precipitation (iv)Filtration (v) Fractional distillation (vi)Saponification	A. Formation of a solid from a solution during a chemical reaction B. Elements with the same atomic number but different mass numbers. C. The process of separating an insoluble solid from a liquid using a filter paper. D. The conversion of a solid directly into a gas without passing through the liquid state. E. A method for separating a mixture of liquids with different boiling points F. The process of making soap from fats and oils. G. The process of separating components based on their differences in mass.

SECTION B (54 MARKS)

Answer all questions in this section.

3. (a) Define the following terms:
   (i) Atomic number
   (ii) Mass number
   (b) The electronic configuration of an element is 2, 8, 2.
   (i) Name the element.
   (ii) To which group and period does this element belong?
   (c) Write the chemical formula of the compound formed between the element in 3(b) and chlorine.

4. (a) Balance the following chemical equations:
   (i) $NaOH + H_2SO_4 \rightarrow Na_2SO_4 + H_2O$
   (ii) $Fe_2O_3 + CO \rightarrow Fe + CO_2$
   (b) Calculate the mass of sodium hydroxide (NaOH) that can be prepared from 46 g of sodium.

5. (a) Salt is one of the compounds that we normally use every day. State its definition and give two examples of the compound.

(b) Explain the difference between a strong acid and a weak acid, giving one example for each.

(c) State three methods of preparing soluble salts.

6. (a) Gas Y is prepared in the laboratory by adding Hydrogen peroxide drop by drop to manganese (IV) oxide, which catalyzes the decomposition of the peroxide, and is collected is collected over water. Name gas Y and draw a labeled diagram showing the laboratory preparation of the gas.

(b)State three uses of gas Y

7.(a) Explain the meaning of electrolysis.

(b) Give three applications of electrolysis.

(c) Describe the electrolysis of concentrated sodium chloride solution using carbon electrodes.

8. (a) The solution obtained after mixing marble chips with sulphuric acid was allowed to evaporate to dryness in the same container. After the evaporation process, the container together with its contents had a total weight of 97.3g. On the next day it was found to have 98.6g.

(i) Why did the mass increase?

(ii) What is the name of the property shown by the substance left in the container?

(iii) What would happen to the rate of reaction if calcium carbonate powder was used instead of marble chips? Explain.

(b) Solution A and B were mixed in one beaker. After a few minutes, the beaker was very hot. As a Chemistry student, explain the phenomena by:

(i) Give the type of the reaction. Explain your answer.

(ii) How would you adjust the temperature in such a reaction to favor the forward reaction

SECTION C (30 MARKS)

Answer any two questions from this section.

9. (a) Define the terms oxidation and reduction in terms of electron transfer.

(b) Using relevant examples, differentiate between oxidizing and reducing agents.

(c) For the reaction:

$Fe+CuSO_4 \rightarrow FeSO_4 + Cu$, identify the substance that is oxidized, the substance that is reduced, the oxidizing agent, and the reducing agent.

10. (a) Explain four factors that affect the rate of a chemical reaction.

(b) Explain what is meant by chemical equilibrium.

(c) With reference to the Haber process ($N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$), explain how a change in pressure affects the yield of ammonia gas.

11. (a) Write the electronic configurations for Magnesium and cholerine
    (b) With the aid of a diagram, explain the formation of an ionic bond between Magnesium and Chlorine.
    (c) Explain the trend in reactivity of group I elements (alkali metals) down the group.